Exercise 2.7: Unbiased Constant-Step-Size Trick

Problem Statement

In most of this chapter we have used sample averages to estimate action values because sample averages do not produce the initial bias that constant step sizes do. However, sample averages are not a completely satisfactory solution because they may perform poorly on nonstationary problems. Is it possible to avoid the bias of constant step sizes while retaining their advantages on nonstationary problems?

One way is to use a step size of

$$\beta_n \doteq \frac{\alpha}{\bar{o}_n}, \tag{1}$$

to process the nth reward for a particular action, where $\alpha > 0$ is a conventional constant step size, and $\bar{o}_n$ is a trace of one that starts at zero:

$$\bar{o}_n \doteq \bar{o}_{n-1} + \alpha(1 - \bar{o}_{n-1}), \qquad n \geq 0, \qquad \bar{o}_0 \doteq 0. \tag{2}$$

Carry out an analysis like that leading to equation (2.6) to show that Q_n is an exponential recency-weighted average without initial bias.

Solution

Consider one fixed action. Suppose that the rewards observed from this action are

$$R_1, R_2, \ldots, R_n.$$

The action-value estimate is updated after observing the nth reward by

$$Q_{n+1} = Q_n + \beta_n(R_n - Q_n), \tag{3}$$

where

$$\beta_n = \frac{\alpha}{\bar{o}_n}.$$

We will show that this produces a normalized exponential recency-weighted average of the observed rewards, with no remaining weight on the arbitrary initial estimate Q_1.

Step 1: Solve the trace recursion

The trace update is

$$\bar{o}_n = \bar{o}_{n-1} + \alpha(1 - \bar{o}_{n-1}).$$

Rewrite it as

$$\bar{o}_n = \alpha + (1 - \alpha)\bar{o}_{n-1}. \tag{4}$$

Starting from $\bar{o}_0 = 0$, unrolling gives

$$\begin{aligned}
\bar{o}_n &= \alpha + \alpha(1-\alpha) + \alpha(1-\alpha)^2 + \cdots + \alpha(1-\alpha)^{n-1} \\
&= \sum_{k=0}^{n-1} \alpha(1-\alpha)^k \\
&= 1 - (1-\alpha)^n.
\end{aligned}$$

Thus,

$$\boxed{\bar{o}_n = 1 - (1-\alpha)^n.} \tag{5}$$

This quantity is the cumulative total weight that ordinary constant-step-size updating would have assigned to the observed rewards after n observations.

Step 2: Substitute the corrected step size

Using $\beta_n = \alpha/\bar{o}_n$ in the update gives

$$\begin{aligned}
Q_{n+1} &= Q_n + \frac{\alpha}{\bar{o}_n}(R_n - Q_n) \\
&= \left(1 - \frac{\alpha}{\bar{o}_n}\right) Q_n + \frac{\alpha}{\bar{o}_n} R_n.
\end{aligned}$$

Multiplying both sides by $\bar{o}_n$ yields

$$\bar{o}_n Q_{n+1} = (\bar{o}_n - \alpha) Q_n + \alpha R_n. \tag{6}$$

From the trace recursion in (4),

$$\bar{o}_n - \alpha = (1-\alpha)\bar{o}_{n-1}.$$

Therefore,

$$\bar{o}_n Q_{n+1} = (1-\alpha)\bar{o}_{n-1} Q_n + \alpha R_n. \tag{7}$$

Now define the scaled estimate

$$S_n \doteq \bar{o}_n Q_{n+1}. \tag{8}$$

Then (7) becomes

$$S_n = (1-\alpha) S_{n-1} + \alpha R_n, \qquad S_0 = \bar{o}_0 Q_1 = 0. \tag{9}$$

Step 3: Unroll the recursion

Unrolling (9), we obtain

$$\begin{aligned}
S_n &= \alpha R_n + (1-\alpha) S_{n-1} \\
&= \alpha R_n + \alpha(1-\alpha) R_{n-1} + (1-\alpha)^2 S_{n-2} \\
&= \alpha R_n + \alpha(1-\alpha) R_{n-1} + \alpha(1-\alpha)^2 R_{n-2} + \cdots + \alpha(1-\alpha)^{n-1} R_1.
\end{aligned}$$

Equivalently,

$$S_n = \sum_{i=1}^{n} \alpha(1-\alpha)^{n-i} R_i. \tag{10}$$

Since $S_n = \bar{o}_n Q_{n+1}$, we divide by $\bar{o}_n = 1 - (1-\alpha)^n$:

$$\boxed{Q_{n+1} = \frac{\sum_{i=1}^{n} \alpha(1-\alpha)^{n-i} R_i}{1 - (1-\alpha)^n}}. \tag{11}$$

Step 4: Interpret the result

Equation (11) is an exponential recency-weighted average. The most recent reward R_n has numerator weight α, the previous reward R_{n-1} has numerator weight $\alpha(1-\alpha)$, the reward before that has numerator weight $\alpha(1-\alpha)^2$, and so on.

However, unlike the ordinary constant-step-size update, these weights are normalized by

$$1-(1-\alpha)^n.$$

Thus the actual weight on reward R_i is

$$w_i = \frac{\alpha(1-\alpha)^{n-i}}{1-(1-\alpha)^n}. \tag{12}$$

These weights sum to one:

$$\begin{aligned}\sum_{i=1}^n w_i &= \frac{\sum_{i=1}^n \alpha(1-\alpha)^{n-i}}{1-(1-\alpha)^n} \\ &= \frac{1-(1-\alpha)^n}{1-(1-\alpha)^n} \\ &= 1.\end{aligned}$$

Therefore, all the weight is placed on the observed rewards $R_1, \dots, R_n$.

Comparison with ordinary constant step size

For ordinary constant-step-size updating,

$$Q_{n+1} = Q_n + \alpha(R_n - Q_n),$$

unrolling gives

$$Q_{n+1} = (1-\alpha)^n Q_1 + \sum_{i=1}^n \alpha(1-\alpha)^{n-i} R_i. \tag{13}$$

The term

$$(1-\alpha)^n Q_1$$

is the initial bias: the estimate continues to depend on the arbitrary initial value Q_1, especially for small n.

In contrast, the corrected update in (11) has no term involving Q_1. Therefore, it keeps the main advantage of constant step sizes - exponentially weighting recent rewards more heavily, which is useful in nonstationary problems - while removing the initial-value bias.

Final Answer

Using

$$\beta_n = \frac{\alpha}{\bar{o}_n}, \qquad \bar{o}_n = 1-(1-\alpha)^n,$$

the estimate after observing n rewards is

$$Q_{n+1} = \frac{\sum_{i=1}^n \alpha(1-\alpha)^{n-i} R_i}{1-(1-\alpha)^n}.$$

This is a normalized exponential recency-weighted average of the observed rewards. Since the weights sum to one and there is no remaining coefficient multiplying Q_1, the estimate has no initial bias.